PAPER_1743_BEAL_GCD_GT_1

Daniel T. Murphy

PAPER_1743 — Beal Conjecture = A^x+B^y=C^z (x,y,z ≥ 3) requires gcd(A,B,C) > 1 ($1M Beal prize)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Number Theory
Date: June 18, 2026
Status: CLOSED — EXACT closure (PAPER_1271-1299 broader corpus)

Observation

PAPER_1296 (PAPER_1271-1299 broader corpus) gives:

```
A^x+B^y=C^z (x,y,z ≥ 3) requires gcd(A,B,C) > 1 ($1M Beal prize)
```

Status: **EXACT**.

UQFF Closed Identity

```
A^x+B^y=C^z (x,y,z ≥ 3) requires gcd(A,B,C) > 1 ($1M Beal prize) EXACT
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1296
- Related: PAPER_1716-1725 (tier-30 Millennium); PAPER_1726-1735 (tier-31 neutron lifetime)
- Calculator dispatch: `calculate_paradox({"paradox": "beal_gcd_gt_1"})`
